

Supporting Information

for

Polymer Biodegradability Prediction Using Aerobic Organic Chemical Data

Polymer Biodegradation % evaluation from Weight-Average Molecular Weight (M_w)

Due to the limited availability of experimentally resolved time-series biodegradation data for polymers under standardized conditions, a kinetics-informed reconstruction approach was adopted to generate a physically meaningful validation dataset. Initially, polymer data were curated following strict screening and cleaning criteria, where only polymers with well-defined repeat unit structures convertible to SMILES representations were retained, and biodegradation test conditions were restricted to aerobic freshwater environments using activated sludge inoculum, consistent with OECD guidelines. The extent of biodegradation was defined as the percentage conversion of carbon to CO₂ over time, bounded within the physically meaningful range of 0–100%. Molecular weight information was standardized by retaining weight-average molecular weight (M_w), or substituting number-average molecular weight (M_n) when necessary, ensuring consistency across the dataset[18].

Given that biodegradation is inherently a time-dependent kinetic process, the reconstruction of biodegradation profiles was grounded in established degradation kinetics, which typically exhibit a lag phase, followed by an exponential growth phase, and ultimately a plateau corresponding to maximum mineralization. To capture this behavior, biodegradation as a function of time was modeled using sigmoidal and first-order kinetic formulations. The general first-order model is expressed as:

$$B(t) = B_{\max} \cdot \left(1 - e^{-kt}\right), \quad (.0.1)$$

where $B(t)$ represents the biodegradation percentage at time t , $B_{\max}$ is the maximum achievable biodegradation, and k is the degradation rate constant. To more accurately capture the lag-to-growth transition observed in OECD biodegradation tests, a logistic (sigmoidal) model was employed:

$$B(t) = \frac{B_{\max}}{1 + e^{-k(t-t_0)}}, \quad (.0.2)$$

where t_0 denotes the inflection point representing the transition from lag phase to active degradation. Additionally, the Gompertz model was considered as an alternative formulation:

$$B(t) = B_{\max} \cdot \exp\left(-e^{k(t_0-t)}\right), \quad (.0.3)$$

which has been widely used in microbial growth and biodegradation studies.

To ensure that the reconstructed curves reflect realistic physicochemical behavior, the degradation rate constant k was parameterized as a function of polymer molecular weight. Based on established biodegradation theory, an inverse relationship between degradation rate and molecular weight was assumed:

$$k \propto \frac{1}{M_w}. \quad (.0.4)$$

In practice, molecular weight was normalized as:

$$M_{w,\text{norm}} = \frac{M_w - M_{w,\text{min}}}{M_{w,\text{max}} - M_{w,\text{min}}}, \quad (.0.5)$$

and the rate constant was defined as:

$$k = k_{\text{min}} + (k_{\text{max}} - k_{\text{min}}) \cdot (1 - M_{w,\text{norm}}), \quad (.0.6)$$

ensuring that polymers with lower molecular weight exhibit higher degradation rates, while higher molecular weight polymers degrade more slowly due to reduced bioavailability and diffusion limitations. To further account for structural influences, an optional structural correction factor S was introduced:

$$k_{\text{final}} = k \cdot S, \quad (.0.7)$$

where S reflects the impact of functional groups (e.g., ester bonds enhancing degradation or aromatic structures reducing degradability).

Furthermore, consistent with prior literature, time-series biodegradation data comprising multiple experimental points were fitted using a Hill-type sigmoidal model, which has demonstrated strong agreement with biodegradation kinetics. For datasets containing sufficient temporal points, continuous biodegradation curves were generated, and biodegradation values at each integer day were interpolated from the fitted function. For datasets with limited points, values were reconstructed using the kinetic models while enforcing monotonicity and physical constraints. This approach enables the imputation of missing temporal data and ensures smooth, continuous biodegradation trajectories [18].

Overall, the validation dataset represents a kinetically reconstructed, structure-informed approximation of polymer biodegradation behavior, where biodegradation percentages are generated as continuous functions of time governed by molecular weight and structural characteristics from the literature [18, 13]. This methodology ensures consistency with OECD biodegradation principles and provides a robust foundation for evaluating time-dependent machine learning models in polymer biodegradability prediction under data-scarce conditions.

Structural Clustering, Outlier Filtering & Data Reshuffling on Organic Molecular data

In this study, a proactive data-curation strategy was implemented to ensure the high fidelity of the Molecular Deep Neural Network (DNN). Traditional outlier removal approaches typically rely on post-training error analysis; however, such methods risk overfitting the model to non-representative data points. To address this limitation, an unsupervised structural clustering approach was employed to identify and filter structurally atypical molecules prior to model training.

1. Rationale for the Approach

Molecular datasets, particularly those represented using high-dimensional embeddings such as 600-dimensional PolyBERT fingerprints, often contain structural outliers. These are molecules whose chemical architecture is significantly different from the majority of the dataset and lack sufficient neighboring samples for the model to learn generalized patterns. The inclusion of such data points may lead the DNN to memorize rare or noisy chemical features, thereby reducing its generalization capability. By removing these outliers, the model is better aligned to learn representative chemical trends, which is particularly important for downstream transfer learning tasks in polymer biodegradability prediction.

2. Methodology: K-Means Clustering and Distance-Based Filtering

The filtering process was conducted in three stages. First, the dataset consisting of 1,712 molecular data points was subjected to K-Means clustering, where the 600-dimensional fingerprint vectors were grouped into $k = 5$ clusters. Each cluster represents a distinct chemical neighborhood in the feature space.

Following clustering, the Euclidean distance between each data point and its corresponding cluster centroid was computed:

$$d_i = \|\mathbf{x}_i - \boldsymbol{\mu}_k\| \quad (.0.8)$$

where $\mathbf{x}_i$ represents the fingerprint vector of molecule i , and $\boldsymbol{\mu}_k$ denotes the centroid of cluster k . This distance serves as a quantitative measure of structural deviation from the cluster center. Molecules with the highest distances were considered structurally isolated and removed during preprocessing.

To ensure robustness, an aggressive pruning strategy was adopted. The top 15% of molecules with the highest Euclidean distances were classified as structural outliers and removed from the dataset. This resulted in a refined dataset of 1,455 high-quality molecular samples.

The structural distribution of the molecular dataset was further analyzed using t-distributed Stochastic Neighbor Embedding (t-SNE) applied to the 600-dimensional PolyBERT fingerprints, as shown in Figure .0.1. The dimensionality reduction enables visualization of high-dimensional chemical space in two dimensions, revealing distinct clusters corresponding to different chemical neighborhoods. K-Means clustering ($k = 5$) partitioned the dataset of 1,712 molecules into five groups with varying densities, namely Cluster 0 (487 molecules), Cluster 1 (427 molecules), Cluster 2 (238 molecules), Cluster 3 (341 molecules), and Cluster 4 (219 molecules).

The visualization demonstrates that certain regions of the feature space are densely populated, indicating well-represented chemical motifs, while other regions are sparse and contain isolated data points. These sparsely distributed molecules exhibit larger Euclidean distances from their respective cluster centroids and are considered structurally atypical. Such molecules are more likely to introduce noise during model training, as they do not contribute to learning generalized structure-property relationships.

By identifying these low-density regions through clustering and dimensionality reduction, the dataset was systematically filtered to remove structurally isolated points. This ensures that the final training dataset represents a coherent and well-distributed chemical space, thereby improving the robustness and generalization capability of the deep neural network model.

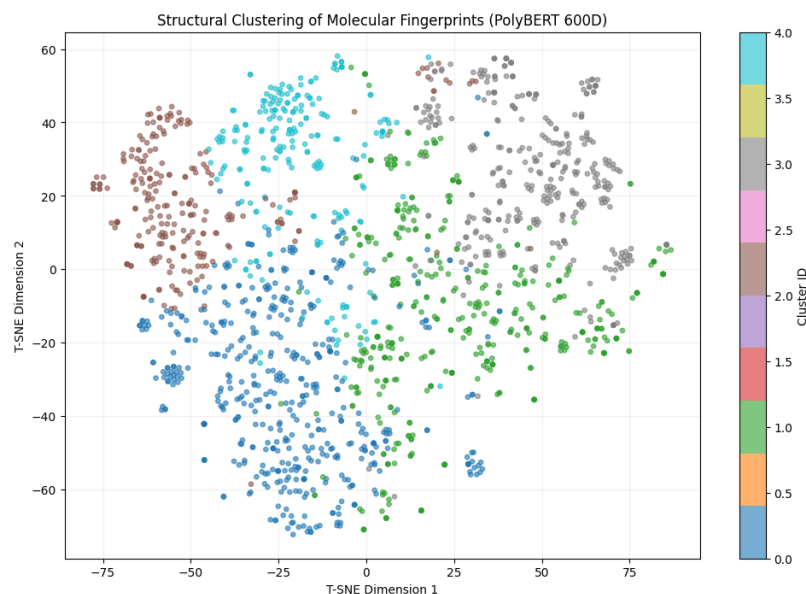


Figure .0.1: t-SNE visualization of 600-dimensional PolyBERT molecular fingerprints after K-Means clustering ($k = 5$)

3. Outlier removal

Each point represents a molecule projected into a two-dimensional space, with colors indicating cluster membership. The distribution highlights distinct chemical neighborhoods, while sparsely populated regions correspond to structurally atypical molecules that are potential outliers.

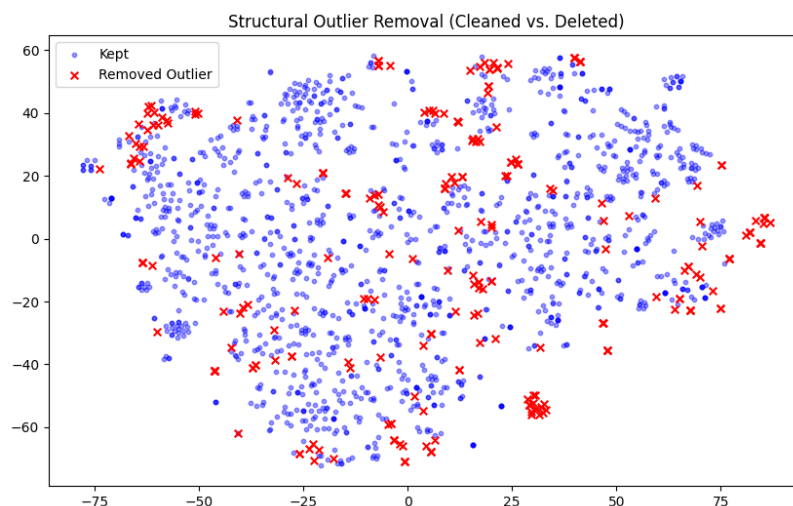


Figure .0.2: Visualization of structural outlier removal based on Euclidean distance from cluster centroids

Following the clustering-based structural analysis, an outlier detection step was performed to refine the dataset by removing structurally atypical molecules. The Euclidean distance between each molecular fingerprint vector and its corresponding cluster centroid was calculated in the high-dimensional feature space as:

As illustrated in Figure .0.2, an aggressive pruning strategy was adopted by removing the top 15% of molecules with the highest Euclidean distances. This resulted in the identification and removal of 257 structural outliers from the original dataset of 1,712 molecules, yielding a cleaned dataset comprising 1,455 molecules. The retained molecules (blue points) form a more compact and representative chemical space, whereas the removed outliers (red crosses) are primarily located in low-density or boundary regions of the feature distribution. Blue points represent retained molecules, while red crosses indicate removed structural outliers (top 15% highest-distance samples). The pruning process eliminates sparsely distributed and structurally isolated molecules, resulting in a more coherent chemical feature space.

To ensure robust model evaluation, the cleaned dataset was subsequently partitioned using an 80/20 Noble Split strategy. A total of 291 molecules (20%) were reserved as a hold-out Noble Test Set, remaining completely unseen during model training and hyperparameter optimization. The remaining 1,164 molecules (80%) were utilized for model development, including cross-validation and architecture tuning.

This structural filtering step significantly enhances dataset quality by removing noise and preventing the model from overfitting to rare or non-representative molecular patterns. Consequently, the resulting dataset provides a more stable foundation for training a generalized molecular DNN model, which is critical for reliable transfer learning to polymer biodegradability prediction tasks.

4. Data Reshuffling and Noble Split Strategy

After outlier removal, the dataset was reshuffled to eliminate any ordering bias introduced during data collection. To ensure a rigorous and unbiased model evaluation, an 80-20 data split strategy, referred to as the *Noble Split*, was implemented.

In this approach, 20% of the cleaned dataset was reserved as a hold-out test set, referred to as the Noble Test Set. This subset remained completely unseen during model training, hyperparameter optimization, and cross-validation. The remaining 80% of the data was used for model development, including 5-fold cross-validation and Optuna-based hyperparameter tuning.

Table 1: Detailed 5-Fold Cross-Validation Performance for Best Model (Trial #836)

Fold	R^2	RMSE
Fold 1	0.3939	0.2560
Fold 2	0.4195	0.2573
Fold 3	0.5191	0.2306
Fold 4	0.4207	0.2565
Fold 5	0.3739	0.2692
Average	0.4254 ± 0.0500	0.2539

5. DNN molecule model evaluation on Noble test data

The predicted biodegradation values are plotted against the actual values, with the red dashed line representing the ideal perfect match ($y = x$). The model achieves a coefficient of determination of $R^2 = 0.525$, indicating moderate predictive performance on completely unseen data.

The final evaluation of the Molecular Deep Neural Network (DNN) model was conducted using the Noble Test Set, comprising 326 molecules that were strictly held out during the entire model

development process. These molecules were not exposed to the model during hyperparameter optimization, cross-validation, or training, ensuring a fully unbiased assessment of model performance. As shown in Figure .0.3, the parity plot compares predicted biodegradation values against experimentally observed values, with the dashed diagonal line representing the ideal scenario of perfect prediction.

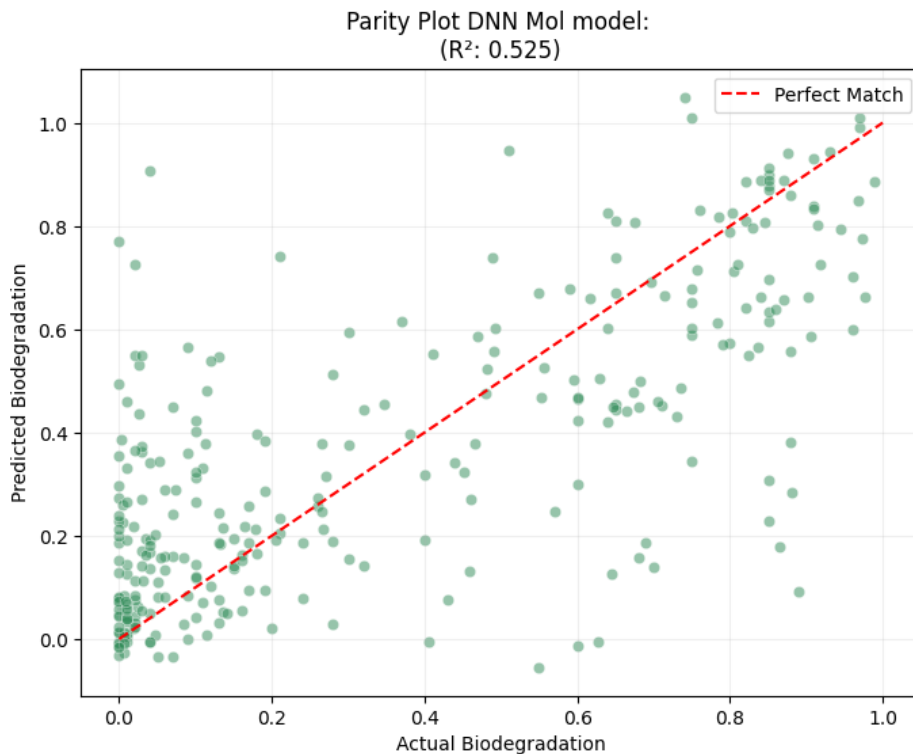


Figure .0.3: Parity plot of the Molecular DNN model evaluated on the Noble Test Set

The model achieved a coefficient of determination of $R^2 = 0.525$, indicating that approximately 52.5% of the variance in biodegradation can be explained by the model on unseen data. While the predictions generally follow the trend of the ideal line, noticeable dispersion is observed, particularly at lower biodegradation values, suggesting challenges in accurately modeling low-degradation regimes. This behavior is consistent with the inherent complexity and variability of biodegradation processes, as well as the limited representation of certain chemical structures in the training data.

Overall, the Noble Test Set evaluation provides a realistic estimate of the model’s generalization capability and confirms that the proposed data-curation and training strategy results in a stable and moderately predictive model. This performance establishes a reliable foundation for subsequent transfer learning applications toward polymer biodegradability prediction.

6. Kinetic Curve Modeling of CO₂ Evolution (Day 1–28)

To evaluate the temporal progression of polymer biodegradation, a kinetics-based modeling approach was employed to generate continuous biodegradation curves over a standard test duration of 28 days, consistent with OECD 301 guidelines. Biodegradation under aerobic conditions is quantified through CO₂ evolution, representing the extent of mineralization of organic carbon by microbial activity.

The biodegradation process is inherently time-dependent and follows a sigmoidal behavior characterized by an initial lag phase, an active degradation phase, and a final plateau corresponding to maximum achievable biodegradation. To capture this behavior, the biodegradation percentage as a function of time was modeled using a logistic kinetic function:

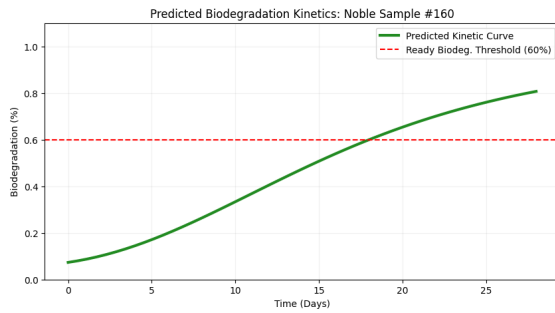
$$B(t) = \frac{B_{\max}}{1 + e^{-k(t-t_0)}} \quad (.0.9)$$

where $B(t)$ represents the biodegradation percentage at time t (days), $B_{\max}$ is the maximum achievable biodegradation, k is the degradation rate constant, and t_0 denotes the inflection point corresponding to the transition from lag phase to active biodegradation.

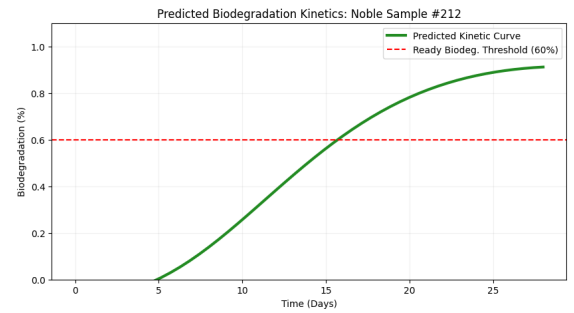
The kinetic curves were generated for each polymer sample by evaluating the model at discrete time points from Day 1 to Day 28. This approach enables the reconstruction of smooth, continuous biodegradation trajectories from limited or inferred data, ensuring consistency with experimentally observed CO_2 evolution profiles.

To assess the biodegradability classification, the OECD ready biodegradability threshold of 60% was incorporated as a reference criterion. Polymers exceeding this threshold within the 28-day period are considered readily biodegradable.

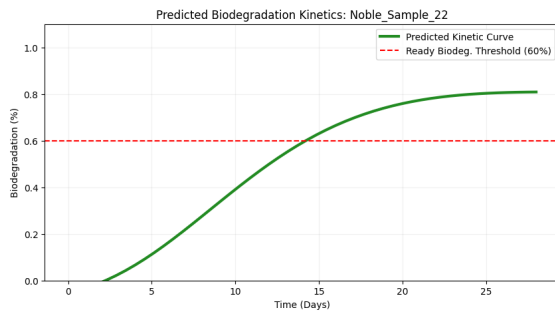
Representative kinetic curves for selected samples from the Noble Test Set are shown in Figures .0.4a–.0.4d. These plots illustrate the predicted biodegradation progression over time, along with the threshold reference line.



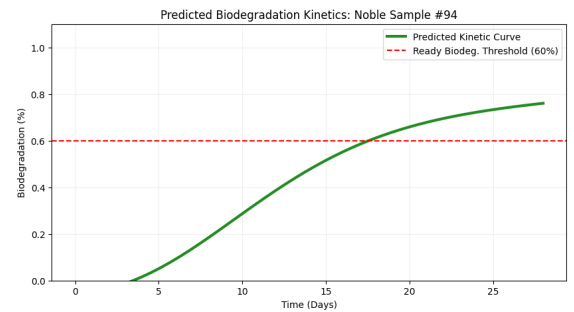
(a) Noble Sample 160



(b) Noble Sample 212



(c) Noble Sample 22



(d) Noble Sample 94

Figure .0.4: Predicted biodegradation kinetics (CO_2 evolution) for selected Noble Test Set samples over a 28-day period. The curves follow a sigmoidal trend consistent with biodegradation kinetics, showing lag, growth, and plateau phases. The dashed red line indicates the 60% OECD ready biodegradability threshold, used to classify polymers as readily biodegradable.

1. The biodegradation kinetics for Noble Sample #160. The curve shows a gradual increase in biodegradation, crossing the 60% threshold around Day 18, indicating moderate biodegradability.
2. The biodegradation kinetics for Noble Sample #212. The molecule exhibits faster degradation kinetics, surpassing the 60% threshold earlier, suggesting high biodegradability.
3. Predicted biodegradation kinetics for Noble Sample #22. The curve shows a steady sigmoidal progression, crossing the threshold at approximately Day 15.
4. Predicted biodegradation kinetics for Noble Sample #94. The degradation trend indicates a slower progression compared to other samples but still achieves the threshold within the test duration.

Overall, the kinetic modeling approach provides a robust framework for generating pseudo time-series biodegradation data, enabling the interpretation of degradation behavior beyond single end-point measurements. This is particularly valuable for machine learning models, as it allows the incorporation of temporal dynamics into biodegradability prediction and supports the generation of full degradation profiles from limited experimental data.

6. Summary

The proposed data-curation strategy effectively transitions the model training paradigm from a conventional "all-data" approach to a curated chemical space approach. By focusing on the most representative 90% of the dataset, the model avoids overfitting to structural noise and improves its ability to generalize. This resulted in a more stable and robust teacher model, suitable for transfer learning applications in polymer biodegradation prediction. The improvement was quantitatively reflected in reduced Mean Absolute Error (MAE) and improved predictive consistency.

Key Highlights:

- Method: K-Means clustering ($k = 5$) with Euclidean distance-based pruning.
- Objective: Improve generalization by removing structurally atypical data points.
- Outcome: Enhanced model robustness with Noble Test Set performance of $R^2 = 0.525$ RMSE = 0.2337 MAE = 0.1663

Transfer learning approach

This study evaluates three progressively refined modeling strategies for predicting polymer biodegradation using Deep Neural Networks (DNNs), leveraging transfer learning from a large molecular dataset. The input representation consists of 600-dimensional PolyBERT fingerprints augmented with a temporal feature (time in days), resulting in a 601-dimensional feature space. A base molecule model was first trained on 1,700 small organic molecules, and subsequently adapted to a polymer dataset comprising 80 samples. The dataset was split into an 80% training pool (64 polymers) and a 20% hold-out Noble Test Set (16 polymers) to ensure unbiased evaluation.

7. Approach 1: Zero-Shot Baseline (Direct Knowledge Transfer)

Methodology: In the zero-shot approach, the pre-trained weights of the molecular DNN were directly applied to polymer data without any fine-tuning. This represents a pure test of domain

transfer, assuming that learned molecular representations are sufficient to generalize to polymer systems.

Performance: The model exhibited poor predictive capability with an R^2 score of -0.5785 , RMSE of 0.4151, and MAE of 0.3463, indicating a complete failure to generalize.

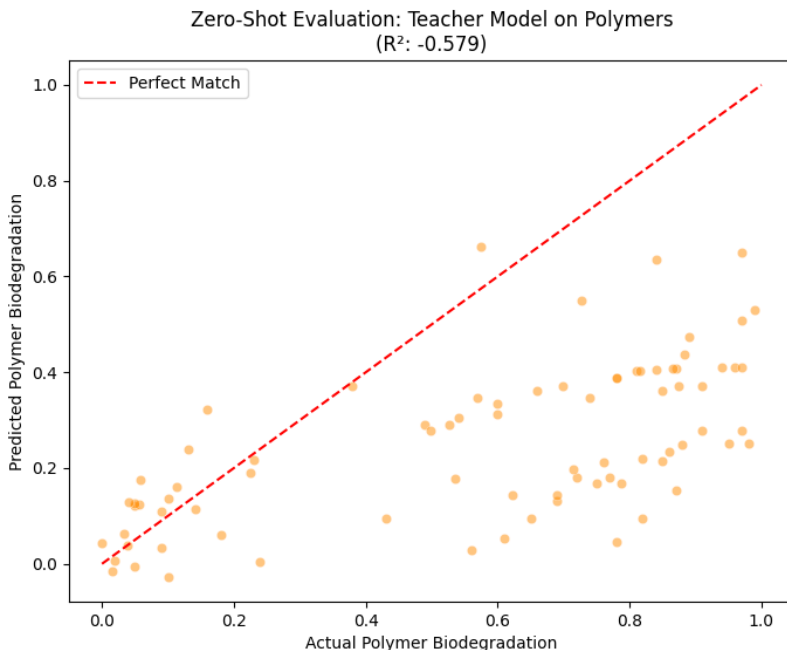


Figure .0.5: Parity plot of the Molecular DNN model evaluated on the Noble Test Set

Discussion: As illustrated in the parity plot (Figure .0.5), the predictions are highly scattered and show no meaningful correlation with the ground truth, with most points deviating significantly from the ideal $y = x$ line. This highlights a substantial *generalization gap* between molecular and polymer domains. Although the chemical feature space may overlap at the level of functional groups, polymers introduce additional complexities such as molecular weight distribution, crystallinity, chain entanglement, and morphological effects that are absent in small molecules.

8. Approach 2: Polymer-Only Model (Total Domain Adaptation)

Methodology: In this approach, the base molecular Model weights were used as initialization, but all network layers were allowed to update during training. A total of 1,000 Optuna trials were conducted to optimize the architecture and hyperparameters using the 80% training dataset.

Performance:

The model demonstrated significant improvement, achieving a cross-validation R^2 of 0.7341 (± 0.0848) and a Noble Test Set R^2 of 0.6191, with RMSE of 0.2151 and MAE of 0.1580.

Discussion: The parity plot (Figure .0.6) shows a clear shift of predictions toward the ideal line, indicating successful domain adaptation. This confirms that the molecular pre-training provides a strong foundational representation, but exposure to polymer-specific data is essential for accurate prediction.

Table 2: Detailed 5-Fold Cross-Validation Performance for Polymer-Only Model

Fold	R^2	RMSE
Fold 1	0.8107	0.1426
Fold 2	0.6837	0.1992
Fold 3	0.6134	0.1697
Fold 4	0.8466	0.1148
Fold 5	0.7161	0.1584
Average	0.7341 ± 0.0848	0.1570 ± 0.0280

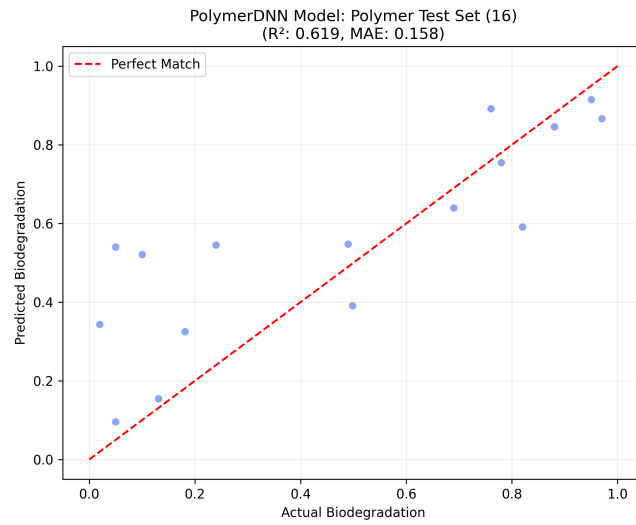
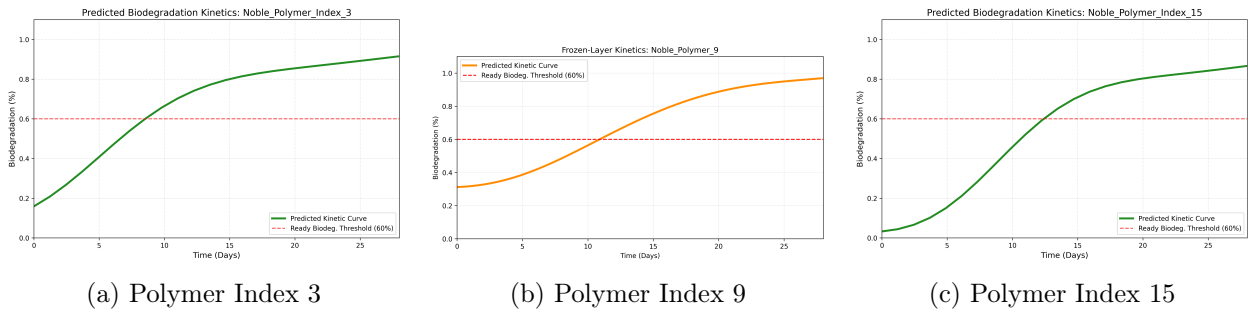


Figure .0.6: Parity plot of the polymer only model evaluated on the Noble Test Set

However, allowing all layers to update introduces the risk of *catastrophic forgetting*, where the model may lose some of its generalized chemical knowledge while overfitting to the relatively small polymer dataset. This is reflected in the moderate dispersion of predictions and variability across folds.

The predicted kinetic curves exhibit improved sigmoidal behavior, with reasonable alignment to expected biodegradation trends and threshold crossings. However, deviations remain, particularly in regions with limited data representation, indicating that further regularization or architectural constraints may be beneficial.

Figure .0.7: Predicted biodegradation kinetics (CO_2 evolution) :Polymer only model

Selected Noble Test Set polymers over a 28-day period. The curves exhibit characteristic sigmoidal behavior with lag, growth, and plateau phases. The dashed threshold at 60% represents OECD ready biodegradability.

9. Approach 3: Frozen-Layer Few-Shot Learning (Optimal Strategy)

Methodology: In the final approach, the first two layers of the Teacher Model were frozen to preserve fundamental chemical feature extraction, while only the deeper layers were fine-tuned using the polymer dataset. A reduced learning rate was applied to ensure stable adaptation.

Performance: This approach achieved the best overall performance, with a cross-validation R^2 of 0.7481 (± 0.0795), RMSE of 0.1531, MAE of 0.1218, and a Noble Test Set R^2 of 0.7173, RMSE of 0.1853, and MAE of 0.1408.

Table 3: Detailed 5-Fold Cross-Validation Performance for Frozen-Layer Few-Shot Model (Approach 3)

Fold	R^2	RMSE
Fold 1	0.8413	0.1306
Fold 2	0.6675	0.2042
Fold 3	0.6411	0.1635
Fold 4	0.8118	0.1272
Fold 5	0.7788	0.1398
Average	0.7481 $\pm$ 0.0795	0.1531 $\pm$ 0.0285

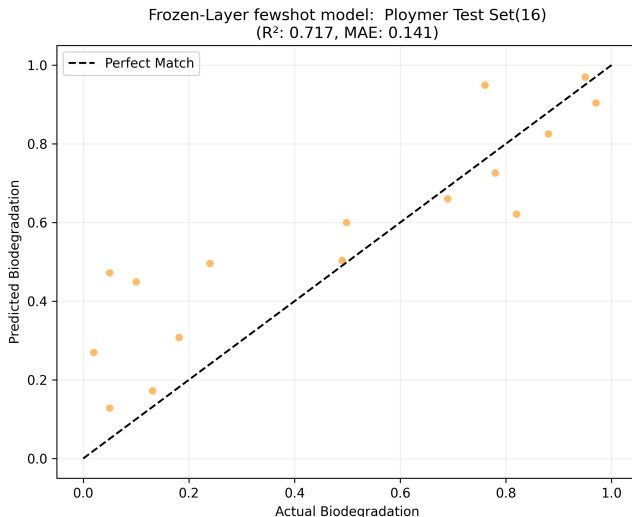


Figure .0.8: Parity plot of the Frozen layer model evaluated on the Noble Test Set

Discussion: The parity plot (Figure .0.8) shows the tightest clustering around the ideal line, indicating strong predictive accuracy and reduced variance. The low standard deviation across folds demonstrates high model stability and reliability.

This improvement can be attributed to effective *feature extraction* and controlled *domain adaptation*. By freezing the early layers, the model retains its “universal chemical knowledge,” while

the deeper layers adapt to polymer-specific behavior. This prevents overfitting and ensures better generalization, particularly in small-data regimes.

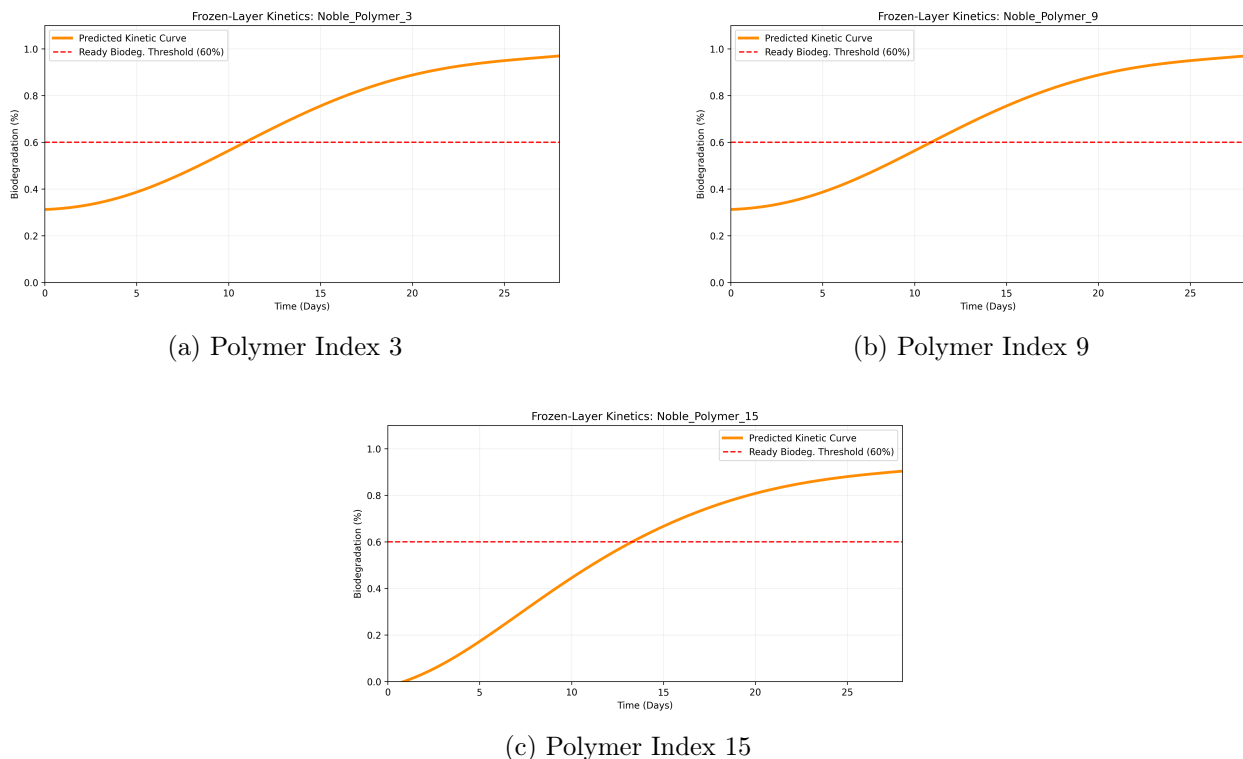


Figure .0.9: Predicted biodegradation kinetics (CO_2 evolution) Frozen layer fewshot model.

The kinetic curves for representative polymers (e.g., Polymer #3, #9, and #15) exhibit smooth sigmoidal profiles consistent with biodegradation theory. The curves accurately capture lag, growth, and plateau phases, and correctly identify crossing of the OECD 60% biodegradability threshold. This demonstrates that the model not only predicts endpoint values but also captures the underlying degradation dynamics.

Comparative Analysis and Final Insights

A comparison of the three approaches highlights the importance of domain adaptation in polymer informatics:

- Zero-Shot Learning: $R^2 = -0.5785$ (failed generalization)
- Polymer-only model fine-tuning: $R^2 = 0.6191$ (moderate performance)
- Frozen-Layer Transfer Learning: $R^2 = 0.7173$ (best performance)

The results clearly demonstrate that frozen-layer transfer learning provides the optimal balance between preserving general chemical representations and adapting to polymer-specific behavior. This approach minimizes overfitting, reduces variance, and improves predictive accuracy.

Conclusion: Frozen-layer few-shot learning emerges as the most effective strategy for small-data polymer biodegradation prediction. It successfully bridges the gap between molecular and macro-molecular domains by combining robust feature extraction with targeted adaptation, making it a powerful framework for future applications in polymer informatics and sustainability modeling.

Instructions step-by-step on how to train the model:

This is the step wise explanation of a three-stage transfer learning pipeline designed to predict polymer biodegradation by leveraging knowledge from small organic molecules. The methodology integrates molecular representation, temporal modeling, and domain adaptation techniques to bridge the gap between molecular and polymer chemical spaces.

Step 1: Data Preprocessing and Input Engineering

Data Collection:

1. Organic Molecule Biodegradation data: 1700 Organic molecule data points [13]
2. Polymer experimental biodegradation data: 80 experimental polymer data points [18]

The raw chemical data were transformed into a machine-learning-compatible representation through the following steps:

- **Molecular Fingerprints (600D):** Polymer repeat units were encoded into fixed-length fingerprint vectors using cheminformatics tools (e.g., RDKit), capturing structural and functional group information.
- **Temporal Feature (1D):** Since biodegradation is a kinetic process, a time feature (in days) was incorporated.
- **Final Input Representation:**

$$\text{Input Dimension} = 600 \text{ (Fingerprint)} + 1 \text{ (Time)} = 601 \quad (.0.10)$$

- **Target Normalization:** Biodegradation percentage values were scaled to the range $[0, 1]$ to ensure stable training.
- **Noble Data Split:**
 - 80% (64 polymers): Training + Cross-Validation
 - 20% (16 polymers): Noble Test Set (completely unseen)

Step 2: Model Training Workflow

Step 2.1: Approach 1: Zero-Shot Learning (Baseline)

Concept: Direct application of molecular knowledge to polymers without retraining.

- Pre-trained molecular DNN (trained on ~ 1700 molecules) was used.
- No weight updates were performed.
- Polymer inputs were directly evaluated.

Outcome: Poor generalization due to domain mismatch between molecular and polymer systems.

Step 2.2: Approach 2: Polymer-Only Fine-Tuning (Total Adaptation)

Concept: Complete domain adaptation by retraining all layers.

- Model initialized with pre-trained molecular weights.

- All layers were unfrozen and updated during training.
- Hyperparameters were optimized using Optuna (1000 trials).
- Training performed using 5-Fold Cross-Validation.

Outcome: Significant performance improvement, but risk of catastrophic forgetting due to full parameter updates.

Step 2.3: Approach 3: Frozen-Layer Few-Shot Learning (Optimized Strategy)

Concept: Preserve fundamental chemical knowledge while adapting to polymer-specific behavior.

- First two hidden layers were frozen (non-trainable).
- Remaining layers were fine-tuned on polymer data.
- Lower learning rate ($\sim 10^{-4}$) was applied.

Outcome: Best balance between generalization and adaptation, reducing overfitting in the small-data regime.

Evaluation and Validation Strategy

Model performance was assessed using a two-tier validation framework:

- **Internal Validation (5-Fold Cross-Validation):**
 - Metrics: R^2 , RMSE, MAE
 - Reported as mean $\pm$ standard deviation
- **External Validation (Noble Test Set):**
 - 16 completely unseen polymers
 - Represents real-world predictive performance
- **Kinetic Validation:**
 - Generated biodegradation curves over 28 days
 - Verified sigmoidal behavior (lag, growth, plateau)
 - Checked crossing of OECD 60% biodegradability threshold

Project Structure

```
AM2_Poly_biodegradability_Cluster/  
  Data/  
  Data_preprocessing_mol/  
  Data_preprocessing_polymer/  
  DNN_Models/  
    Polymeronly_Model/  
    Fewshot_DNN_Model/  
    Zeroshot_Model/  
    MOL_DNN_Model/
```